

CardioTwin AI

Real-Time Cardiometabolic Risk Screening for Nigeria

"From Data to Prevention: AI as Your Health Partner"

 7,750 Hardware cost per unit	4 sec To first live score	5 langs Supported languages	0-100 CardioTwin Score range
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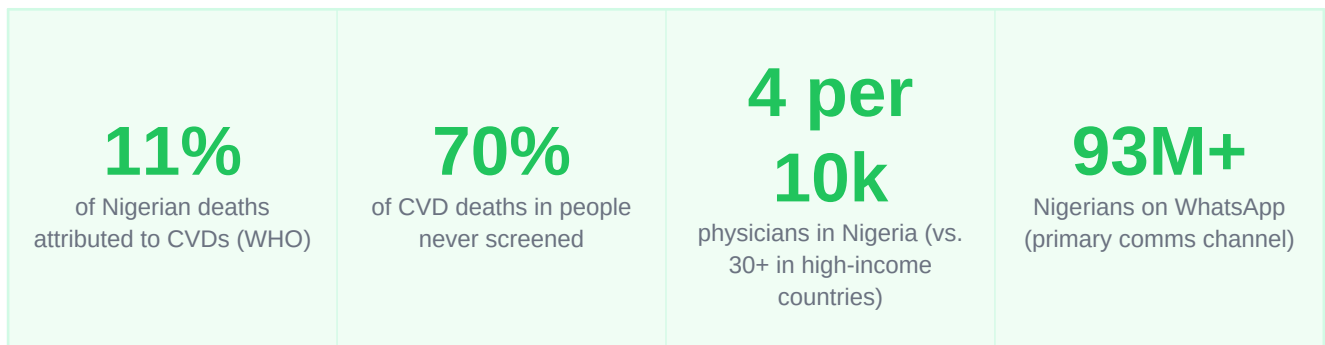
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SECTION 01

Problem Statement

Cardiovascular disease (CVD) is the silent crisis hiding inside Nigeria's health system. Despite being responsible for roughly 11% of all deaths nationally, the vast majority of CVD fatalities occur in people who were **never screened**. The gap between risk and awareness is not a medical problem — it is a systems problem.



The core failure is immediacy. A person's body responds to stress, exertion, and poor lifestyle in real time — but the feedback loop with their health is delayed by months, clinics, queues, and costs. By the time a diagnosis arrives, the window for easy prevention is often closed.

Our hypothesis: If people can see a real-time, personalized number that visibly reacts to what their body is doing right now, they will make better decisions — because consequences become immediate, not abstract.

"Every 33 seconds, someone dies of cardiovascular disease. In Nigeria, 7 out of 10 of those people were never screened."

Solution Overview

CardioTwin AI is a station-based real-time cardiometabolic screening system that gives anyone a personalised health score in under 30 seconds — with no app download, no hospital visit, and no medical training required.

The Three-Layer System

SENSE

An ESP32 microcontroller with MAX30102 optical sensor and NTC thermistor captures heart rate, HRV, SpO₂, and skin temperature every 2 seconds when a finger is placed on the station.

SCORE

A cloud-hosted AI engine processes the raw biometrics through personalised baseline calibration, weighted composite scoring, anomaly detection, and safety gating — producing the CardioTwin Score (0–100) and a health zone in real time.

NUDGE

Alerts and wellness guidance are delivered instantly via the web dashboard, WhatsApp (where 93 million Nigerians already communicate), SMS fallback, and a physical buzzer on the station. Caregiver and medical professional contacts receive automatic alerts when readings cross critical thresholds.

What Is In Scope

- Working hardware station with optical finger sensor
- Real-time biometric streaming to cloud backend every 2 seconds
- CardioTwin Score algorithm (0–100) with 4 health zones
- Live web dashboard with 3D health avatar, score chart, and action guidance
- WhatsApp + SMS alert delivery via Twilio
- What-if risk projection (90-day lifestyle scenario simulation)
- Manual biometric entry fallback when hardware is unavailable
- Two-tier emergency contacts: caregiver (family) + medical professional
- Multi-language support: English, Pidgin, Yoruba, Igbo, Hausa
- Safety guardrails: red-flag detection, LLM output validation, alert throttling

What Is Out of Scope (Pitched as v2 Vision)

- Wearable device form factor — the algorithm is built; the form factor evolves
- Mobile native app — web PWA approach covers mobile responsively

- User authentication and multi-user accounts
- Medical-grade diagnostic claims — wellness screening only
- Deep learning models (not needed for this scope; opens future roadmap)

The CardioTwin Score

The CardioTwin Score is the single most important feature of the product. It transforms complex, multi-dimensional biometric data into one number that **anyone can understand at a glance** — regardless of medical background.

Score Formula

CardioTwin Score = (HRV × 0.40) + (Heart Rate × 0.25) + (SpO2 × 0.20) + (Skin Temp × 0.15)

Component Weights & Rationale

HRV — Heart Rate Variability (40%)	The most powerful non-invasive predictor of cardiac autonomic nervous system health. Low HRV is clinically associated with elevated CVD risk, stress, and poor recovery.
Heart Rate (25%)	Direct indicator of cardiovascular load. Elevated resting HR above personal baseline signals physiological strain even before other markers shift.
SpO2 — Blood Oxygen (20%)	Safety gate for respiratory distress and circulatory compromise. Readings below 92% trigger immediate escalation regardless of other scores.
Skin Temperature (15%)	Correlates with autonomic stress response, inflammation, and thermoregulatory function. Complements the other three metrics for a complete picture.

Health Zones

	Zone	Score	Label	Alert Behaviour
■	GREEN	80–100	Thriving	No alert — continue monitoring
■	YELLOW	55–79	Mild Strain	WhatsApp wellness nudge
■	ORANGE	30–54	Elevated Risk	WhatsApp + physical buzzer alert
■	RED	0–29	Critical Strain	WhatsApp urgent + buzzer + caregiver/doctor alert

Calibration Process

The engine runs a personalised baseline calibration on the first 5 readings of each session. This establishes the user's own resting heart rate, HRV, SpO₂, and temperature as the reference point — so every subsequent score is relative to *that individual*, not a population average. The baseline confidence level adjusts alert severity during early calibration to prevent false alarms before a reliable personal baseline is established.

System Architecture

CardioTwin AI is a three-layer cloud-connected system. Each layer is independently deployable and communicates via a well-defined JSON API contract.

Layer	Description
Hardware Layer	ESP32 microcontroller + MAX30102 + NTC thermistor. Collects raw biometrics, validates readings against backend bounds, and POSTs JSON to the cloud API every 2 seconds via WiFi.
Cloud Backend	Python FastAPI deployed on DigitalOcean / Azure. Receives readings, orchestrates the AI engine pipeline, stores sessions in PostgreSQL (SQLite for local dev), and returns structured responses.
AI Engine	Pure Python module — validation → baseline calibration → scoring → zone classification → anomaly detection → safety gating → nudge generation. 410+ unit tests, 100% module coverage.
Notification Layer	Twilio WhatsApp + SMS. Caregiver and medical professional contacts receive automatic alerts when escalation thresholds are crossed.
Web Dashboard	React 19 + TypeScript + Vite + Tailwind CSS 4 + Three.js. Fully responsive (mobile + desktop). Renders live score, 3D health avatar, action guidance, trend chart, what-if projections, and manual entry.

Data Flow (end-to-end)

1. Finger placed on CardioTwin Station → MAX30102 and thermistor begin sampling
2. ESP32 firmware collects 100 samples, computes smoothed BPM, SDNN-based HRV, SpO₂, and skin temp
3. Client-side validation checks values against backend bounds before sending
4. HTTP POST to /api/reading every 2 seconds with JSON payload including source field
5. FastAPI backend passes reading to AI engine pipeline
6. Engine runs: validation → signal quality gate → baseline calibration → CardioTwin Score → zone classification → red-flag check → anomaly detection → nudge generation
7. Structured response returned: status, score, zone, action_summary, safety, signal_quality, disclaimer
8. Dashboard updates in real time; alerts sent via WhatsApp/SMS if thresholds crossed
9. Caregiver/medical professional automatically notified at CONTACT_CAREGIVER, SEEK_HELP, or EMERGENCY escalation levels

Hardware

The CardioTwin Station is intentionally a **station-based design**, not a wearable. The MAX30102 PPG sensor requires firm, stable skin contact for accurate readings — a constraint that wearables mitigate with complex motion-compensation algorithms that took Apple and Garmin years to develop. Our station gives clinical-grade accuracy every time with █7,750 of off-the-shelf components.

Bill of Materials

Component	Purpose	Est. Cost
ESP32-WROOM-32	Microcontroller + WiFi/Bluetooth SoC	~█3,500
MAX30102 Sensor	Heart rate, HRV, SpO█ (optical PPG)	~█2,000
10kΩ NTC Thermistor	Skin temperature (Steinhart-Hart equation)	~█300
10kΩ Resistor	Voltage divider for thermistor ADC circuit	~█50
Mini Vibration Motor (3V)	Tactile alert on health zone change	~█500
2N2222 NPN Transistor + 1kΩ R	Motor driver circuit	~█100
Breadboard + Jumper Wires	Assembly and prototyping	~█800
Micro-USB Cable	Power and programming	~█500
TOTAL		~█7,750 (~\$10 USD)

Firmware Architecture

- **Sensor sampling:** 100-sample windows from MAX30102 at 200Hz sample rate. Maxim proprietary algorithm computes SpO█ and heart rate from RED/IR ratio.
- **Signal smoothing:** 6-reading BPM moving average to reduce beat-to-beat noise.
- **HRV computation:** SDNN (Standard Deviation of NN intervals) from a rolling 12-reading RR-interval buffer — a validated short-term HRV metric.
- **Temperature:** NTC thermistor with Steinhart-Hart Bcoefficient equation, averaged over 20 ADC samples, with a +4°C calibration offset for finger-surface-to-core estimation.
- **Finger detection:** IR signal threshold (>50,000 counts) gates all readings — no finger = no send.
- **Client-side validation:** Values checked against backend bounds (BPM 30–220, HRV 0–200, SpO█ 70–100, temp 30–42°C) before any HTTP POST.

- **Response handling:** Firmware parses calibrating / scored / retake_requested responses and drives green/red LEDs and buzzer accordingly.

Physical Feedback System

Signal	Meaning
Green LED blink	Successful data send to backend
Red LED solid	No finger detected
Red LED + buzzer (3×)	Health alert triggered (ORANGE/RED zone)
Double red blink	Network send failure
Double ascending beep	WiFi connected at startup

AI Engine & Safety System

The AI Engine is a modular Python pipeline with 410+ unit tests and 100% coverage on all core modules. It is designed to be medically conservative — every output passes through a safety layer before reaching the user.

Processing Pipeline

Stage	Description
1. Validation	Input bounds checking and sanitisation of all biometric values.
2. Signal Quality Gate	Rates reading as good/ok/poor with a confidence score. Poor-quality readings return a retake request before any scoring occurs.
3. Baseline Calibration	Personalises resting HR, HRV, SpO ₂ , and temperature from the first 5 readings of the session.
4. Scoring	Computes per-component scores (0–100) and weighted CardioTwin Score.
5. Zone Classification	Maps score to GREEN/YELLOW/ORANGE/RED with urgency metadata.
6. Red-Flag Detection	Checks absolute physiological limits (SpO ₂ <88%, HR >150, temp >39.5°C, etc.) independently of baseline — triggers escalation if crossed.
7. Anomaly Detection	Detects sudden drops, sustained declines, and zone downgrades against personal baseline. Alert throttling prevents spam (max 5 alerts/30 min).
8. Nudge Generation	Calls Groq LLM (llama-3.3-70b) with strict safety constraints. LLM output is validated against 16 blocked diagnostic phrases before delivery. Falls back to multilingual templates if API is unavailable or output is unsafe.
9. Action Summary	Confidence-aware structured guidance with: status, why, next_step, if_symptoms, advice_strength (full/cautious/retake_only), signal_quality, and up to 2 reason drivers.

Safety Guardrails

All system outputs pass through the safety module before reaching the user. This is a non-negotiable layer — it cannot be bypassed.

- **Red-flag thresholds:** 9 absolute physiological limits that escalate independently of baseline (SpO₂ emergency, critical, warning; HR very high/very low/high; temperature very high/very low/high; HRV

critically low).

- **Escalation tiers:** NONE → SELF_CARE → MONITOR → CONTACT_CAREGIVER → SEEK_HELP → EMERGENCY. Each tier carries a specific safe next step and disclaimer.
- **LLM output validation:** 16 blocked phrases scanned in every Groq response (e.g., 'you have', 'heart attack', 'take aspirin', 'diagnosed with'). Any blocked phrase triggers immediate fallback to template.
- **Alert throttling:** Same alert type: max once per 10 minutes. Any alerts: max 5 per 30 minutes. Critical/emergency: never throttled.
- **Baseline confidence gating:** Alert severity is downgraded when the baseline is uncertain (low: 2 levels down, medium: 1 level down). Prevents false alarms during early calibration.
- **Signal quality gating:** Poor-quality readings are rejected before scoring. Users are prompted to retake with proper sensor placement.
- **Disclaimers:** Every response includes a contextually appropriate disclaimer. The system never claims to diagnose or prescribe.

Two-Tier Emergency Contact System

At session start, users optionally provide two contacts: a **caregiver** (family member) and a **medical professional** (doctor/nurse). The system automatically dispatches WhatsApp alerts to the appropriate tier based on escalation level:

Contact Tier	Triggered At	Message Content
Caregiver	CONTACT_CAREGIVER, SEEK_HELP, EMERGENCY	Receives zone, score, and safe next step. Warm, personal language.
Medical Professional	SEEK_HELP, EMERGENCY only	Receives clinical summary with red-flag list. Urgent framing.

Software Stack & APIs

Technology Stack

Layer	Technology
Hardware Firmware	Arduino C++ (ESP32 Arduino core), ArduinoJson, Maxim SpO ₂ algorithm library
Backend Framework	Python 3.12 + FastAPI + SQLAlchemy ORM + Pydantic v2
AI / ML Libraries	NumPy, SciPy, scikit-learn (baseline calibration, projection)
LLM Integration	Groq API — llama-3.3-70b-versatile (async, with safety validation)
Database	PostgreSQL (production) / SQLite (local dev)
Notifications	Twilio WhatsApp + SMS API
Frontend	React 19 + TypeScript + Vite 7 + Tailwind CSS 4
3D Rendering	Three.js + @react-three/fiber + @react-three/drei (health avatar)
Charts	Recharts (score history, trend analysis)
Internationalisation	Custom i18n context with 5-language translation keys
Deployment	DigitalOcean App Platform (backend) + Vite build (frontend)
CI/CD	GitHub Actions + Docker + Procfile

Core API Endpoints

Endpoint	Purpose
POST /api/session/start	Start session with optional user phone, caregiver, and medical professional contacts.
POST /api/reading	Hardware reading submission (every 2s). Returns calibrating, retake_requested, or scored + action_summary.

POST /api/reading/manual	Manual biometric entry when hardware is unavailable. Accepts partial readings with safe defaults.
GET /api/score/{session_id}	Latest score for frontend polling. Includes action summary.
GET /api/history/{session_id}	Full score history for Recharts trend chart.
POST /api/predict	90-day what-if risk projection with optional lifestyle scenario.
GET /api/nudge/{session_id}	On-demand LLM wellness nudge for display and WhatsApp delivery.
POST /api/feedback	User feedback on alerts: helpful / not_helpful / false_alarm.
POST /api/alert	Manual WhatsApp or SMS dispatch.

Live Demo Script

The demo runs for approximately 3 minutes and is structured in four acts. A pre-recorded backup video must exist before the pitch.

ACT 1 — The Hook (30 seconds)

Every 33 seconds, someone dies of cardiovascular disease. In Nigeria, 7 out of 10 of those people were never screened. What if your body could warn you — not your doctor, not a hospital — but your body, in real time, before it's too late? Today we're going to show you exactly that.

ACT 2 — Live Hardware Demo (90 seconds)

A volunteer places their finger on the CardioTwin Station.

- Dashboard shows data streaming live → calibration progress bar fills → Score: ■ 86 — Thriving
- Volunteer performs 30 seconds of physical exercise (burpees or jumping jacks)
- Volunteer returns and places finger on sensor again
- Score drops live: ■ 41 — Elevated Risk
- Vibration motor buzzes on the station
- WhatsApp message arrives on volunteer's phone in real time
- Show action guidance panel: 'Stop activity, rest, hydrate, and retake shortly'

ACT 3 — What-If Projection (30 seconds)

Navigate to the Projection panel. Enter scenario: 'Start regular exercise and reduce sugar.'

- System projects: current 41 → projected 67 over 90 days
- Framing: 'This is not a diagnosis. This is a preview of what consistent action could do.'

ACT 4 — The Vision (30 seconds)

Today: one person, one station, one score — built in 24 hours for ■7,750 in hardware. Tomorrow: community health workers across Nigeria each carrying a CardioTwin device, screening entire villages in hours. Population-level dashboards helping state health ministries allocate resources where risk is highest. From data, to prevention, to partnership. This is CardioTwin AI — your heart's early warning system.

■ Backup Plan

A full demo video must be recorded the night before the pitch. If the hardware station fails at any point, play the video immediately without acknowledging the issue. The audience will not know the difference. Never demo without a working backup.

What-If Risk Projections

The projection engine allows users to simulate how lifestyle changes would affect their CardioTwin Score over a 90-day window. This is the most compelling hook for behaviour change — it makes the future concrete and personal.

Supported Scenarios

Lifestyle Change	Score Impact	Clinical Rationale
Reduce sugar intake	+8.5 points at 90 days	Improving metabolic health reduces cardiovascular load
Quit smoking	+15.0 points at 90 days	Largest single lifestyle impact on CVD risk
Reduce alcohol	+6.0 points at 90 days	Lower alcohol reduces blood pressure and HR
Reduce salt / sodium	+5.0 points at 90 days	Directly reduces hypertension risk
Start regular exercise	+12.0 points at 90 days	Most impactful positive lifestyle factor
Improve sleep	+7.0 points at 90 days	Sleep quality strongly correlates with HRV recovery
Increase water intake	+4.0 points at 90 days	Hydration supports cardiac output efficiency
Eat more vegetables / fibre	+6.5 points at 90 days	Reduced inflammation and improved metabolic markers
Meditation / mindfulness	+5.5 points at 90 days	Stress reduction directly improves HRV

Impact is scaled linearly against $\text{time_factor} = \min(\text{days} / 90, 1.0)$ — so a 45-day projection shows 50% of the stated impact. Score is capped at 100. All projections carry the disclaimer: *'Statistical projection only. Not a medical diagnosis.'*

SECTION 10

Multi-Language Support

CardioTwin AI speaks the languages of Nigeria's people. All wellness nudges, zone alerts, and UI labels are fully translated across five languages, with culturally appropriate tone and phrasing.

Language	Code	Context
English	en	Default — formal, clinical-adjacent
Nigerian Pidgin	pcm	Warm, colloquial — broadest reach across regions
Yoruba	yo	South-west Nigeria — 36M+ speakers
Igbo	ig	South-east Nigeria — 24M+ speakers
Hausa	ha	North Nigeria — 63M+ speakers

Every combination of health zone × language has at least two unique fallback template messages (40 total). Alert-specific messages (SpO2 critical, sudden HRV drop, rapid HR increase, zone downgrade, sustained decline) are also translated across all five languages — ensuring the system works completely offline without the Groq LLM.

Example Nudges by Language

Language × Zone	Message
English (GREEN)	■ You're doing great! Your vitals look excellent. Keep up the healthy habits!
Pidgin (YELLOW)	■ Small attention dey needed. Try rest small and breathe well well.
Yoruba (ORANGE)	■ Ara r■ nílò ìsinmi. Dúró ní ibi ìdák■r■r■r■, mí ■■mí jìn-ín fún ì■■■jú mārùn-ún.
Igbo (RED)	■ Biko kw■s■ ihe niile, zuru ike ugbu a. ■ b■r■ na ■ gagh■ aka mma, kp■■ mmad■.
Hausa (ORANGE)	■ Jiki na bukatar hutu yanzu. Ka/ki tsaya ka/ki huta, yi numfashi a hankali.

Business Model & Scalability

Revenue Streams

Stream	Detail
B2C — Freemium	Free: basic real-time score. Premium: detailed insights, history, what-if, multi-session tracking. Target price: ■2,000/month. Target users: health-conscious urban Nigerians, diaspora.
B2B — Corporate Wellness	White-label CardioTwin stations for corporate wellness programmes, gyms, and pharmacies. Dashboard aggregation for HR/wellness managers. Partners: Leadway Health, AXA Mansard, Corporate HR platforms.
B2G — Government & NGO	State health ministry tools for population-level CVD screening. Community health worker kits (CHW programmes). Population risk heatmaps for resource allocation. Partners: NHIA, state ministries of health, WHO Nigeria.

Unit Economics (Hardware)

■7,750 Hardware cost per CardioTwin unit	~\$10 USD equivalent (accessible at scale)	500+ People screened per unit per day by CHW	■8k target Retail unit price (1× margin at launch)
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Scale Thesis

A community health worker with one CardioTwin unit can screen an entire village in a single day. At ■7,750 per unit and a cloud backend that scales horizontally, the marginal cost of reaching the next 10,000 people is essentially zero. The data generated creates a real-time national CVD risk map — something Nigeria has never had.

Data Privacy & Ethics

Health data is the most sensitive category of personal data. CardioTwin is built with privacy as a first-class design constraint, not an afterthought.

Principle	Implementation
Encryption	TLS 1.3 in transit. Azure/DigitalOcean managed encryption keys at rest.
Regulatory Compliance	Follows NDPR (Nigeria Data Protection Regulation) — Nigeria's primary data protection framework.
Explicit Consent	Users opt in before any data collection begins. Session data is tied to session ID, not personal identity.
No Diagnosis Claim	Every response carries a wellness disclaimer. The phrase 'medical diagnosis' never appears without 'not a' before it. The LLM is constrained from making diagnostic statements.
Data Ownership	Users can export or delete all session data. No data is sold to third parties.
LLM Safety	Groq LLM output is validated against 16 blocked diagnostic phrases before reaching the user. The system prompt explicitly forbids diagnosis, medication recommendations, and disease naming.
Alert Feedback	Users can mark alerts as helpful, not helpful, or false alarm. This feedback improves future alert quality.

Differentiators & Competitive Edge

Why CardioTwin AI Wins

Differentiator	Why It Matters
Working hardware prototype	90% of hackathon teams are software-only. A physical station that works live is the single most credible signal of execution ability.
CardioTwin Score (0–100)	One number that anyone understands instantly. No medical background required. Fitbit gives you data; we give you a decision.
WhatsApp-first delivery	Alerts go to where 93 million Nigerians already live. Zero app download friction. Works on any phone.
■7,750 hardware cost	Proves accessibility at community health worker scale. Not a luxury device — a public health tool.
Five Nigerian languages	English, Pidgin, Yoruba, Igbo, and Hausa. Designed for Nigeria, not adapted from a Western product.
Honest AI framing	We call it an 'Adaptive Risk Model', not a 'Digital Twin' or 'AI Doctor'. Honest claims build trust with judges and, more importantly, with users.
Safety-first architecture	Non-bypassable safety layer: red-flag detection, LLM validation, alert throttling, escalation tiers. Shows engineering maturity beyond the happy path.
Manual entry fallback	If hardware fails during the pitch or in the field, users can type in their readings. Resilience is a feature.
Clear business model	B2C + B2B + B2G revenue paths identified. Unit economics calculated. Shows the team is thinking beyond the hackathon.

Judge Q&A; Preparation

Q: Is this medically accurate?

CardioTwin is a wellness screening tool, not a medical device. Our MAX30102 sensor provides research-grade PPG data, and our scoring algorithm is based on published clinical correlations between HRV and cardiovascular risk. We clearly disclaim this in the UI — every response carries the statement: 'This is a wellness screening tool, not a medical diagnosis.' Our safety layer actively prevents the AI from making diagnostic statements.

Q: Why a station and not a wearable?

Clinical-grade PPG requires firm, stable skin contact. Even Apple Watch uses motion-compensation algorithms that took years to develop. Our station gives accurate data every time with ₦7,750 of hardware. The wearable is our v2 — the algorithm is already built. Form factor is an engineering problem we've deliberately deferred.

Q: How is this different from a Fitbit?

Fitbit gives you data. We give you a decision. The CardioTwin Score collapses complex biometrics into one actionable number, and our nudge system delivers guidance through WhatsApp — where 93 million Nigerians already live. No app download required. Fitbit also doesn't alert your doctor when your SpO₂ drops below 90%.

Q: Can this scale?

The hardware costs ₦7,750. The backend runs on cloud infrastructure that scales horizontally. A community health worker can screen 500 people in a day with one unit. The data aggregates into population-level CVD risk heatmaps for state health ministries — something Nigeria has never had. Marginal cost of reaching the next 10,000 people is essentially zero.

Q: What happens when there's no internet?

The sensor still works locally and all readings are queued. The dashboard falls back to template-based nudges in all five languages — no internet required for the core safety messages. When connectivity resumes, the session syncs automatically.

Q: How do you handle false alarms?

We have three mechanisms. First, baseline confidence gating: alert severity is downgraded during early calibration when personal baselines are uncertain. Second, alert throttling: same alert type maximum once per 10 minutes, maximum 5 alerts per 30 minutes. Third, user feedback: every alert has a 'helpful / not helpful / false alarm' button. This feedback improves the system over time.

Q: What's your data privacy approach?

TLS 1.3 in transit, encrypted at rest, NDPR-compliant. We collect biometric readings tied to session IDs — no name, no national ID required for the basic product. Users can delete all their data at any time. Health data is never sold.

Q: Why Groq and not GPT-4 or Claude?

Groq gives us sub-second LLM inference — critical when we're processing a reading every 2 seconds. Latency matters more than benchmark performance for real-time wellness nudges. And we have safety validation that catches any unsafe output before it reaches the user, regardless of which model we use.

Risk Register & Contingencies

Risk	Prob.	Impact	Mitigation
WiFi fails at venue	High	Critical	Test venue WiFi 2 hours before pitch. Bring mobile hotspot as backup. If both fail: play pre-recorded demo video immediately.
MAX30102 gives erratic readings	Medium	High	Station-based design (not wearable) ensures stable sensor contact. Calibrate with real users the morning of the pitch. Keep spare sensor + 15-min solder time available.
Backend deployment fails	Low	High	Deploy and smoke-test by Hour 8. Have a local laptop backend as fallback. Frontend has offline template fallback built in.
Twilio WhatsApp fails	Medium	Medium	Pre-join Twilio sandbox. Test with real number night before. Fallback: show SMS or mock notification screenshot.
LLM (Groq) API down	Low	Low	Full multilingual template fallback already implemented and tested. The system works completely offline for nudges.
Demo volunteer has unusual readings	Low	High	Pre-screen two volunteers the morning of the pitch. Know their baseline scores. Pick the one with more typical readings.
Judges challenge 'AI' claim	Medium	Medium	Use precise language: 'Adaptive Risk Model with LLM-powered guidance'. Never say 'AI Doctor' or 'Digital Twin'. Be specific about what the ML does.
Hardware enclosure fails visually	Medium	Low	Have a clean backup enclosure assembled. Even a neat breadboard with clear labelling projects competence.

You're not building a product in 24 hours. You're building a story backed by a working prototype. The story wins. The prototype proves you can deliver.

*This document is a living reference — update it as the build progresses.
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